Hydroclimatic Predictions using Machine Learning: Understanding GCMs & CMIP6 Datasets

Bridging Climate Science and Data-Driven Modelling

Weather and/or Climate: Condition of atmosphere i.e.,

Wind, Pressure, Temperature, Humidity, Precipitation, …

- **Weather:** Characteristics of the atmosphere over a short period of time, in terms of few days or weeks. It is the **day-to-day** state of the atmosphere in a region, and its **short-term** (minutes to weeks) variation

  Ex: ***Current*** Temperature, Rainfall, Humidity, Solar radiation

- **Climate:** The statistics of weather over a long period of time. **Climate** is the variation of weather at a given place for a **long time interval**

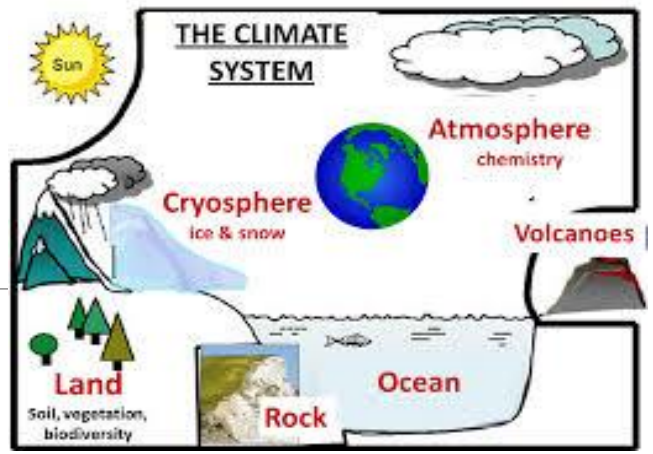
  Ex: ***Average*** Temperature, Rainfall, Humidity, Solar radiation

- **Climate Change:** Difference between climatic states, statistical description over a period of time, altered patterns.

IIIT, HYDERABAD

Earth's Climate

Earth's climate is not static, it changes in response to both natural and anthropogenic forcing.



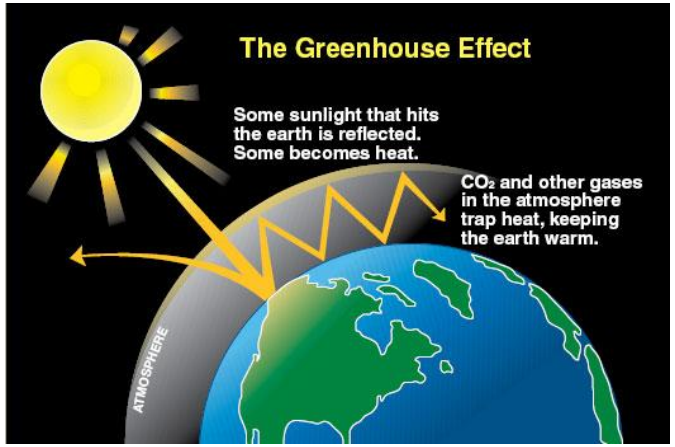
'Atmosphere, hydrosphere, biosphere and geosphere and their interactions' - UNFCCC

Changes in the Atmosphere:

Natural processes

- Volcanic eruption
- Solar radiation fluctuations
- Ocean circulation changes

- Anthropogenic Processes
  - Release of "Greenhouse Gases" into the atmosphere
  - Deforestation



Changes in climatological variables have severe impact on global hydrological cycle affecting the severity and occurrence of water and climate related hazards such as floods, droughts, heat and cold waves etc.

Climate Change –Implications

Increasing Temperatures

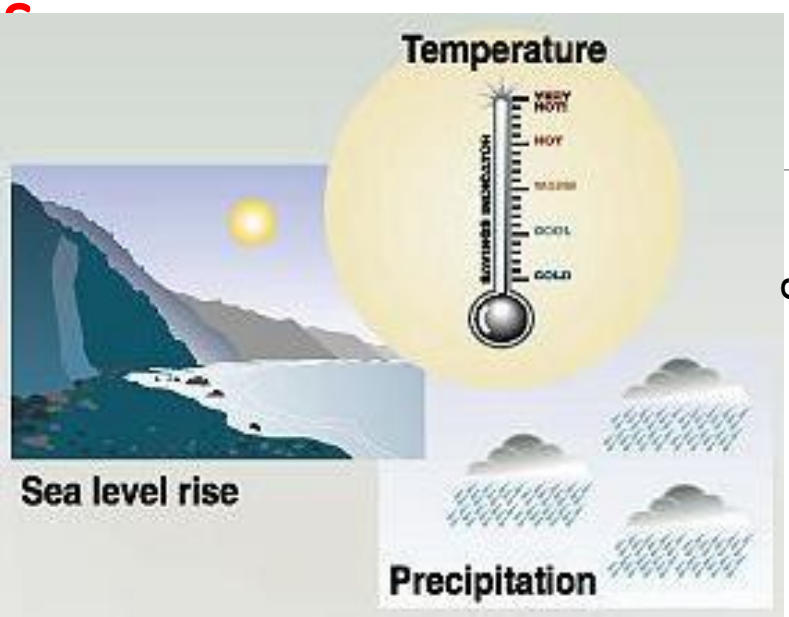
- Evapotranspiration
- Water Quality

Change in Precipitation Patterns

- Streamflow; Water availability
- Intensity, Frequency and Magnitude of Floods and Droughts
- Groundwater Recharge

Rise in Sea Levels

- Inundation of coastal areas
- Salinity Intrusion



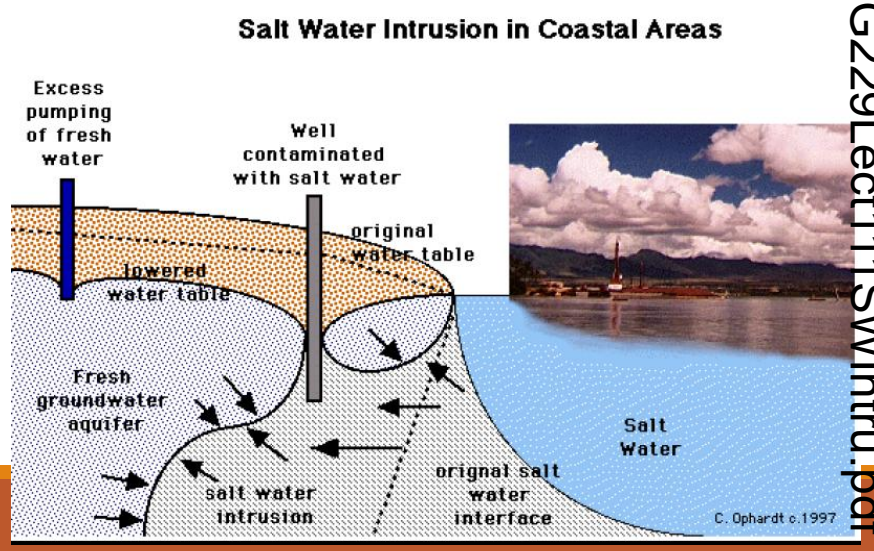
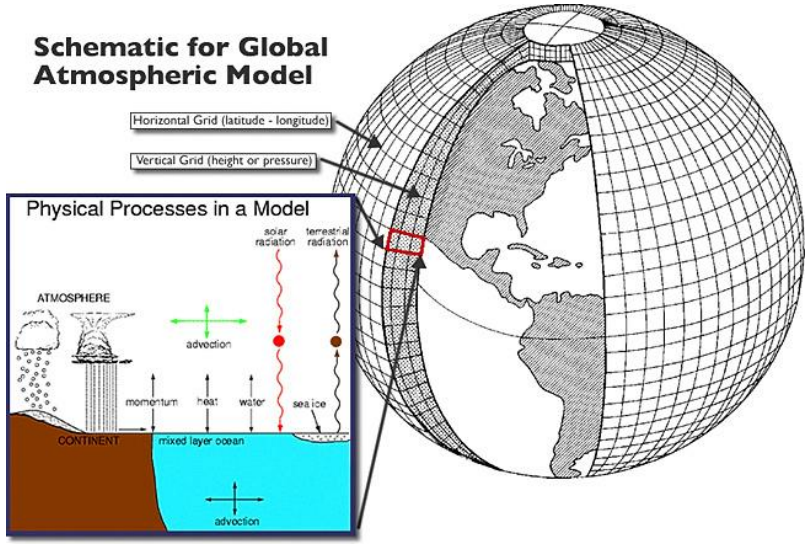


Fig. Source: ww.engr.uconn.edulanboG229Lect11SWIntru.pdf

Climate Models

Entire Earth is divided into Grid cells
(A rectangular area that represents a portion of the Earth's surface)

- They represent the physical, ocean, cryosphere and land surface processes in the atmosphere
- Used to explore past climate events and to project future warming events
- Many climate models have been developed to perform climate projections, i.e. to simulate and understand climate changes in response to the emission of greenhouse gases and aerosols.
- A global, three-dimensional computer model of the climate system that can be used to simulate human-induced climate change. GCMs represent the effects of such factors as reflective and absorptive properties of atmospheric water vapor, greenhouse gas concentrations, clouds, annual and daily solar heating, ocean temperatures, and ice boundaries.



Climate Models or Global Circulation Models (GCMs)

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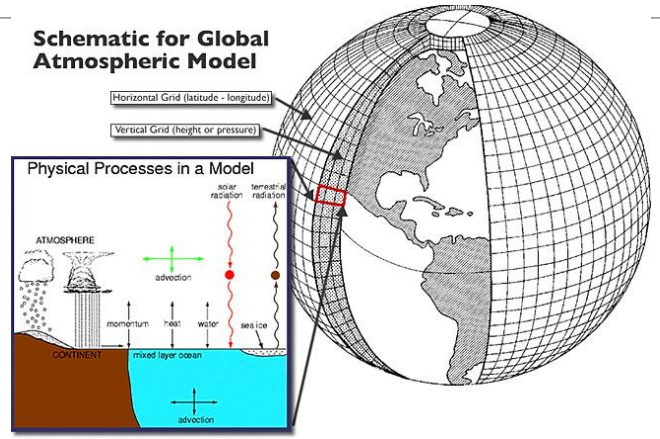
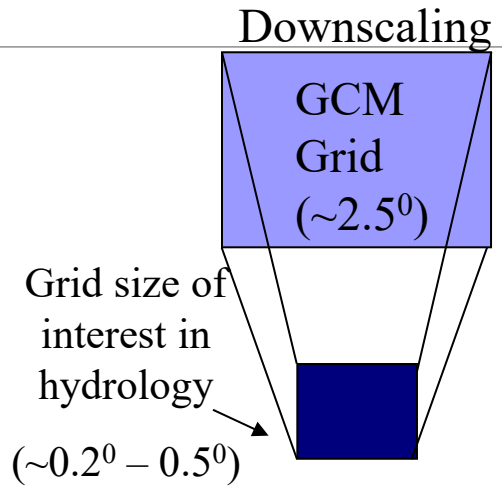
General Circulation Models





- Tools to simulate time series of climate variables globally, accounting for the green house gases (using possible future GHG scenarios) in the atmosphere.
- Can simulate large scale circulation patterns (pressure, wind etc.).
- Cannot produce non-smooth fields such as precipitation.

Hydroclimatic variables (precipitation, temperature, streamflow etc.) are **non-stationary**

- Influenced by:
  - Climate change
  - Land-use change
  - Anthropogenic activities
- Need for:
  - **Future projections**
  - **Data-driven predictive modelling**

Machine Learning + Climate Model Outputs = Powerful Framework

What are Climate Models?

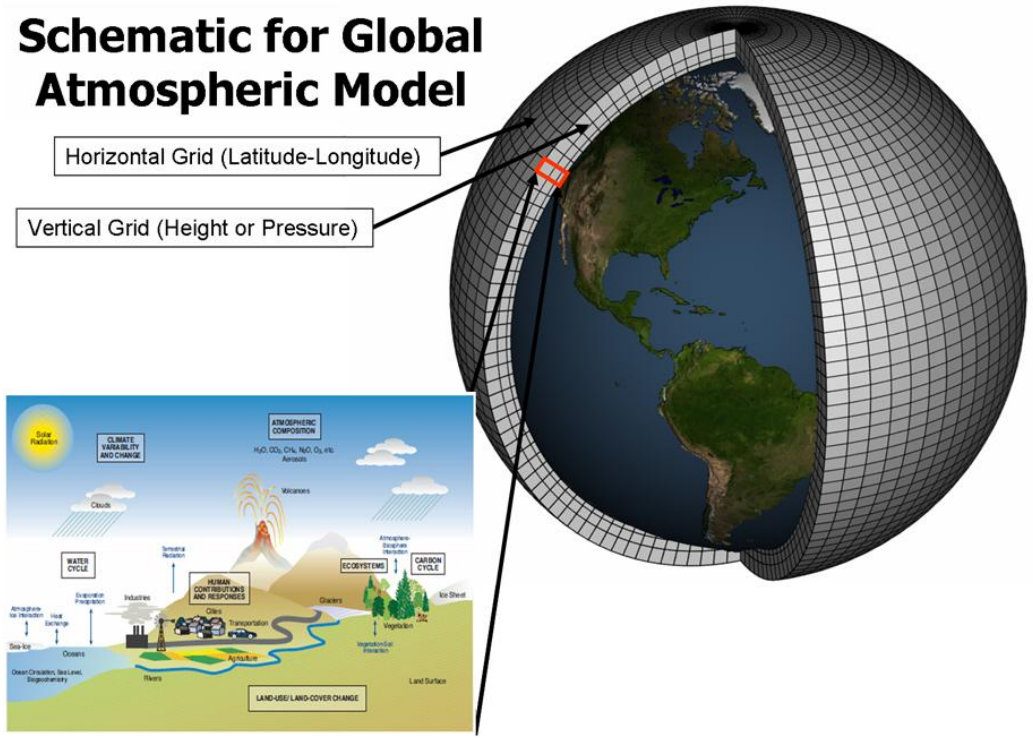
Climate models are complex computer simulations that use mathematical equations based on physics, chemistry, and fluid motion to represent the Earth's climate system.

Climate models simulate Earth’s climate system using:

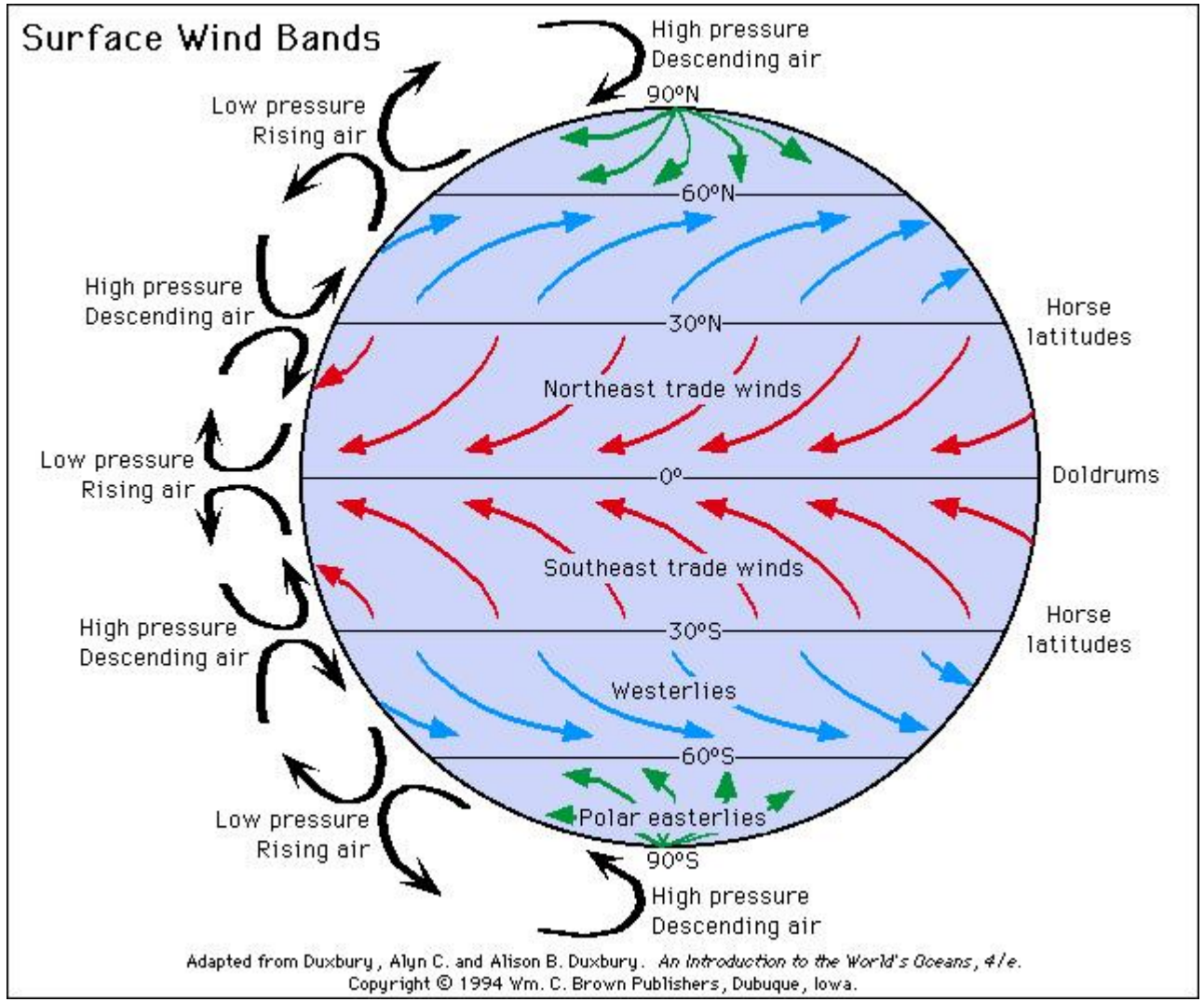
- Physics (fluid dynamics, thermodynamics)
- Chemistry (carbon cycle, aerosols)
- Radiation balance

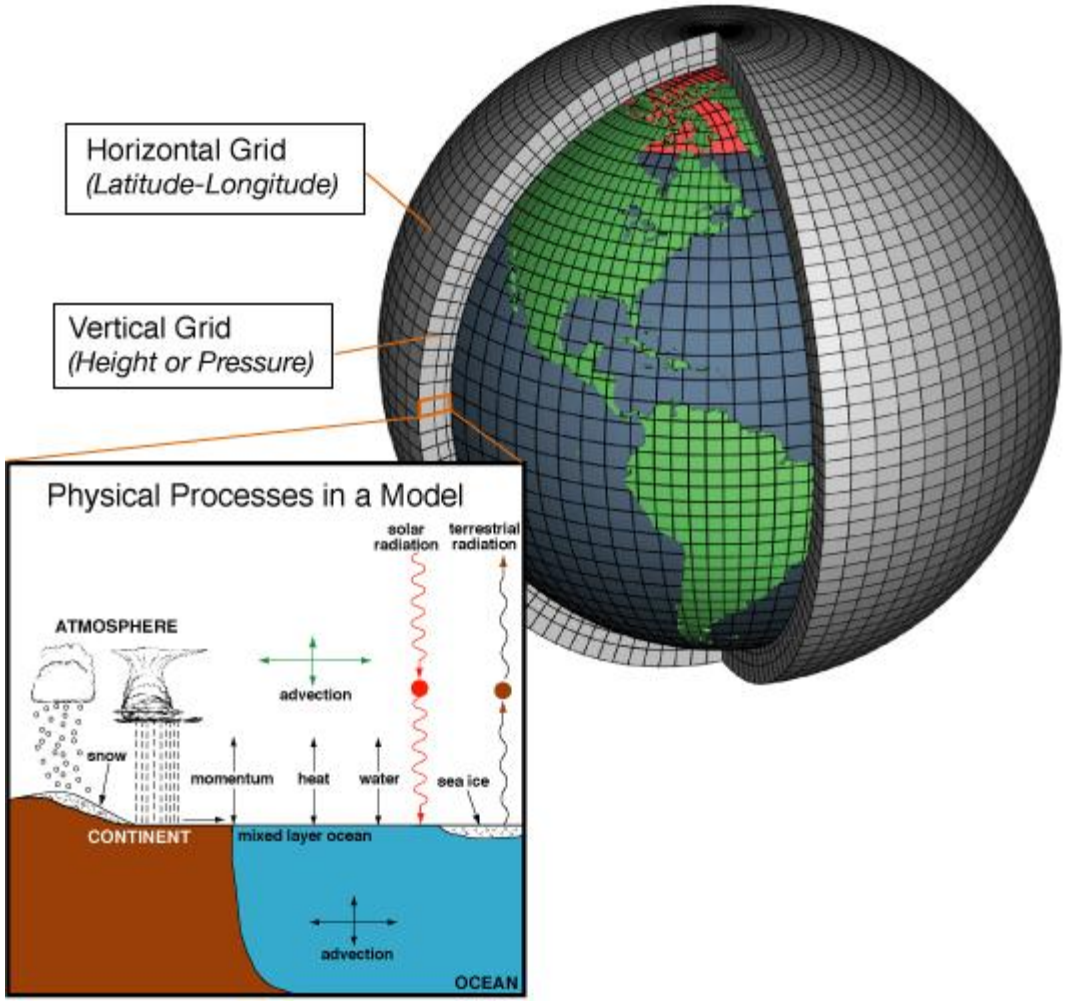
Types:

- Energy Balance Models (EBMs)
- Earth System Models (ESMs)
- **General Circulation Models (GCMs)**

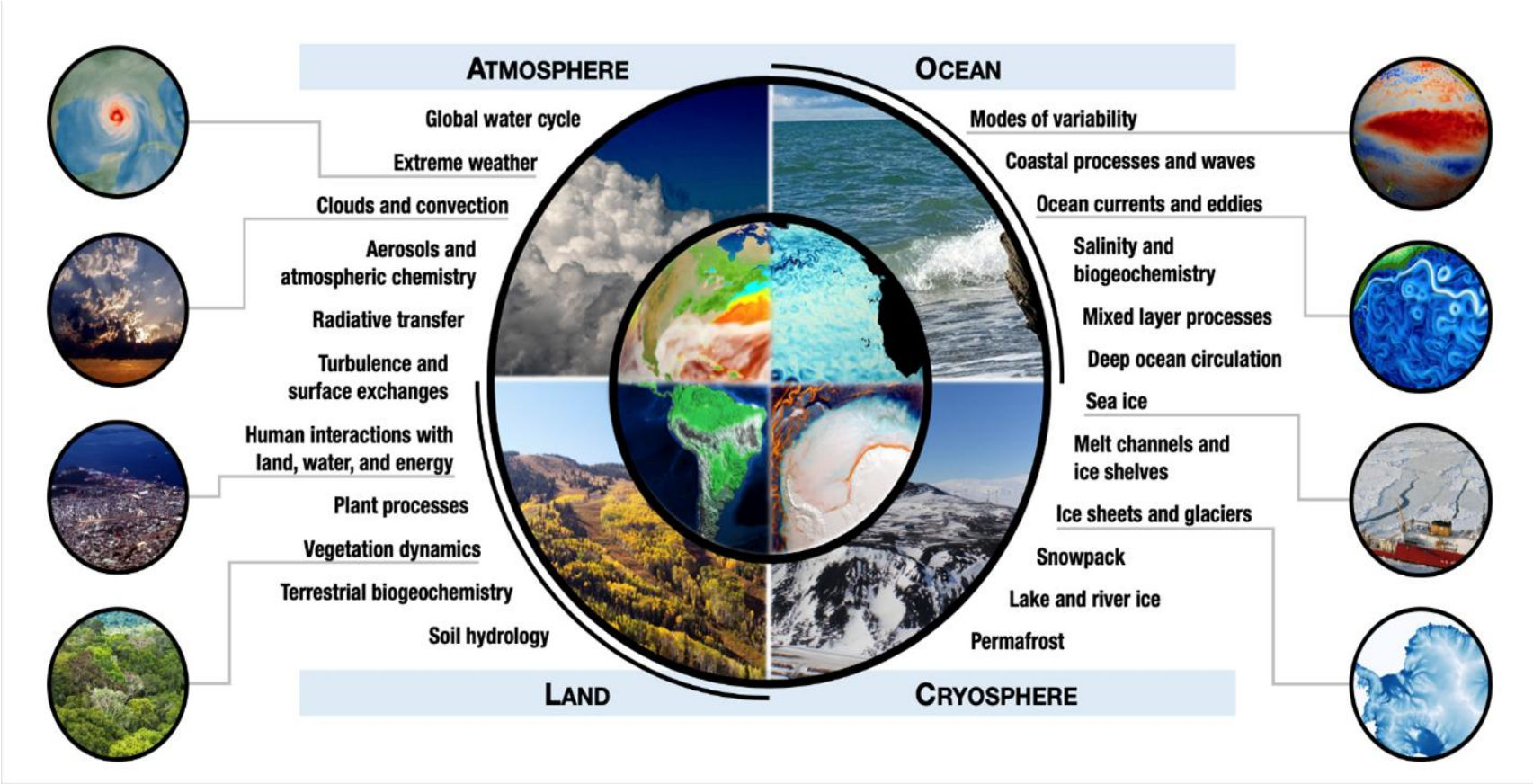


General Circulation Models (GCMs)





General Circulation Models (GCMs)



3D numerical models representing:

- Atmosphere
- Oceans
- Land surface
- Cryosphere

Solve governing equations:

- Navier-Stokes equations
- Energy balance equations

Output Variables:

- Precipitation (pr)
- Temperature (tas, tasmax, tasmin)
- Wind (uas, vas)
- Radiation (rsds)

Limitations of GCMs

- Coarse spatial resolution (~100-250 km)
- Biases in:
  - Rainfall intensity
  - Extremes
- Computational constraints
- Regional applicability is limited

Need for Downscaling

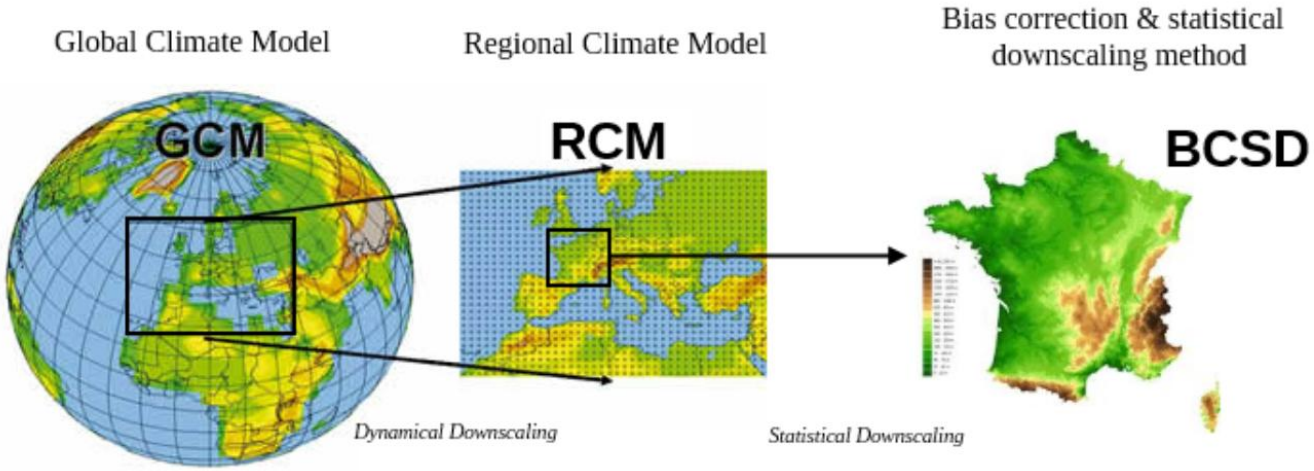
Downscaling Concepts

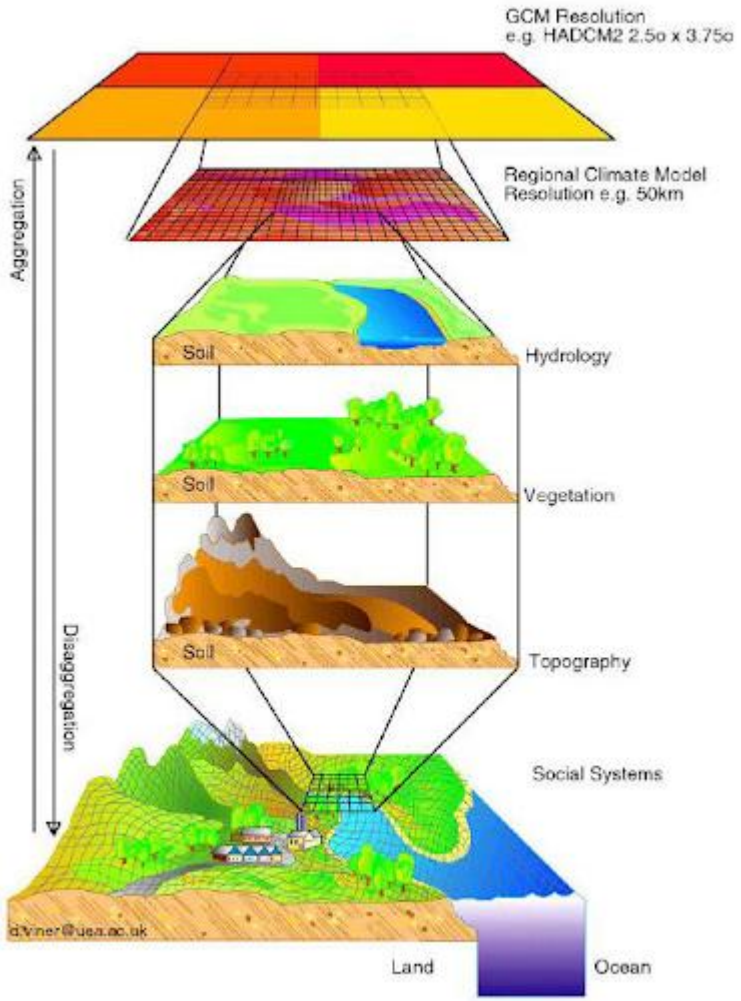
Two approaches:

- Dynamical Downscaling (RCMs)
- Statistical Downscaling

Bias Correction:

- Adjust GCM outputs to match observed climatology



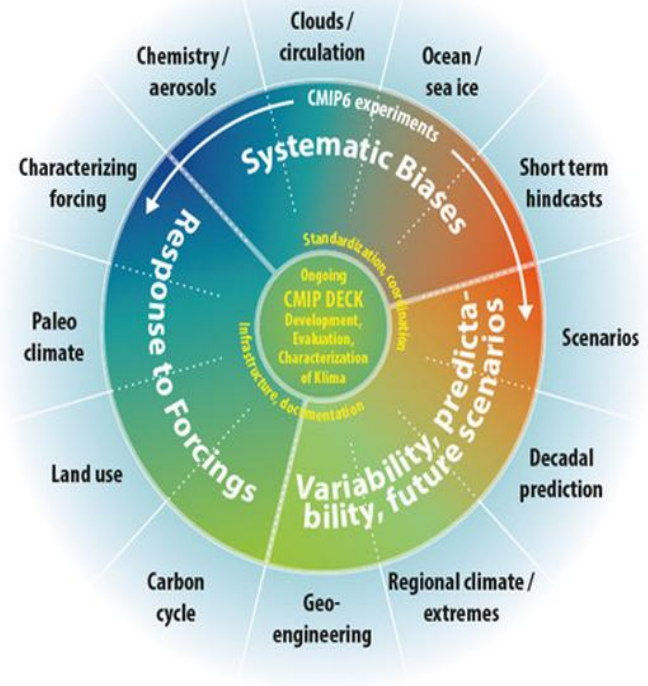


CMIP6 Framework

- Coupled Model Intercomparison Project Phase 6
- Coordinated global effort:
  - Multiple climate modeling centers
- Provides:
  - Standardized experiments
  - Multi-model ensembles

Scenarios:

- SSP1-2.6 (Low emissions)
- SSP2-4.5 (Intermediate)
- SSP5-8.5 (High emissions)



DECK experiments form the core; many specialized MIPs for smaller communities, some 24 of which have been endorsed by CMIP panel. Figure courtesy Meehl et al (*Eos* 2014).

DECK experiments stand for Diagnostic, Evaluation and Characterization of Klima (Klima is Greek for climate).

CMIP6 Variables

Category	Variables
Temperature	tas, tasmax, tasmin
Precipitation	pr
Radiation	rsds
Humidity	hurs
Wind	uas, vas

These serve as **Predictors in ML models**

What is NEX-GDDP CMIP6?

- NASA Earth Exchange Global Daily Downscaled Projections (NEX-GDDP-CMIP6)

Primary Data Variables

- **tasmax**: Daily Maximum Near-Surface Air Temperature (K)
- **tasmin**: Daily Minimum Near-Surface Air Temperature (K)
- **tas**: Daily Mean Near-Surface Air Temperature (K)
- **pr** (or precipitation): Daily Precipitation Rate ($kg \cdot m^{-2} \cdot s^{-1}$)
- **hurs**: Near-Surface Relative Humidity ($\%$)
- **huss**: Near-Surface Specific Humidity (1)
- **rsds**: Surface Downwelling Shortwave Radiation ($W \cdot m^{-2}$)
- **rlds**: Surface Downwelling Longwave Radiation ($W \cdot m^{-2}$)
- **sfcWind**: Daily-Mean Near-Surface Wind Speed ($m \cdot s^{-1}$)

Table: Key Technical Specifications of NEX-GDDP CMIP6 datasets

Feature	Description
Spatial Resolution	0.25° (~25 km)
Temporal Resolution	Daily
Period	1960–2014 (Historical), 2015–2100 (Projections)
Models	~30+ GCMs
Scenarios	SSP1-2.6, SSP2-4.5, SSP3-7.0, and SSP5-8.5

Key Features:

- Bias-corrected
- Statistically downscaled
- Daily resolution

Formats:

NetCDF (.nc)

Perfect for ML **predictors**

Assignment 3
Hydroclimatic variable prediction using Machine Learning

1. Select the hydro climatological variables required to predict (Selection of Predictands)
2. Select the input variables (Predictors) required to predict the Predictands (Coming from Step 1)
3. Develop a training and testing machine learning model with historical data of Predictors (Step 2 Variables) and Predictands (Step 1 Variables).
4. Use the Global Climate Data (e.g. NASA Earth Exchange Global Daily Downscaled Projections) of predictors (Step 2 Variables) to predict the future projections of Predictands (Step 1 Variables).
5. Make inferences about the hydroclimatic variables predicted for the selected case study in terms of visualisations, plotting, etc.

Last date for the Assignment submission is **April 7th, 2026**